Exam Review paper

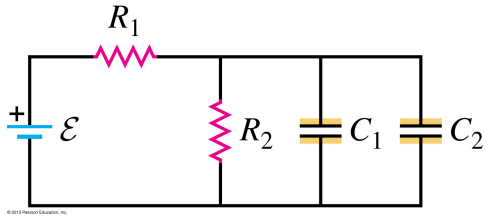
1. A straight 2.00-m, 150-g wire carries a current in a region where the earth's magnetic field is horizontal with a magnitude of 0.55 gauss. (a) What is the minimum value of the current in this wire so that its weight is completely supported by the magnetic force due to earth's field (assuming no forces other than gravity acts on it)? Does it seem likely that such a wire could support this kind of current? (b) Show how the wire would have to be oriented relative to the earth's magnetic field to be supported in this way.

2. A coil 4.00 cm in radius, containing 500 turns, is placed in a magnetic field that varies with time according to $B = (0.0120)\,t + (3.00 \times 10^{-5})\,t^4$. The coil is connected to a 600 Ω resistor, and its plane is perpendicular to the magnetic field. You can ignore the resistance of the coil. (a) Find the magnitude of the induced emf in the coil as a function of time. (b) What is the current in the resistor at time t = 5.00 s?

3. A singly charged Lithium ion has a mass 1.16×10^{-26} kg. It is accelerated through a potential difference of 220 V and then enters a magnetic field with magnitude 0.723 T perpendicular to the path of the ion. What is the radius of the ion's path in the magnetic field?

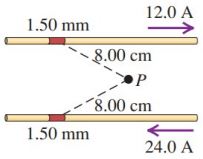
4. Singly ionized atoms are accelerated and then passed through a velocity selector consisting of perpendicular electric and magnetic fields. The electric field is 155 V/m and the magnetic field is 0.0315 T. The ions next enter a uniform magnetic field of magnitude 0.0175 T that is oriented perpendicular to their velocity. (a) How fast are the ions moving when they emerge from the velocity selector? (b) If the radius of the path of the ions in the second magnetic field is 17.5 cm, what is their mass?

5. Consider the circuit shown in the below figure. The battery has emf 60.0 V and negligible internal resistance. $R_2 = 2.00\ \Omega$, $C_1 = 3.00\ \mu F$ and $C_2 = 6.00\ \mu F$. After the capacitors have attained their final charges, the charge on C_1 is $Q_1 = 18.0\ \mu C$. (a) What is the final charge on C_2? (b) What is the resistance R_1?



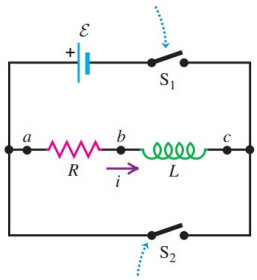
6. A point charge $q_1 = -4.00$ nC is at the point (0.600, 0.800) and a second point charge $q_2 = +6.00$ nC is at the point (0.600, 0). Calculate the magnitude and direction of the net electric field at the origin due to these two point charges.

7. Two parallel wires are 5.80 cm apart and carry currents in opposite directions, as shown in the below figure. Find the magnitude and direction of the magnetic field at point P due to two 1.50-mm segments of wire that are opposite each other and each 8.00 cm from P.

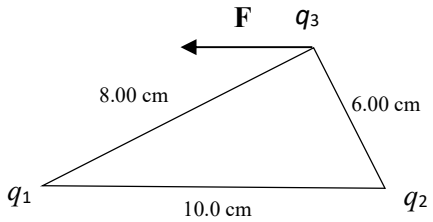


8. Two long, parallel wires are separated by a distance of 2.30 cm. The force per unit length that each wire exerts on the other is 3.80×10^{-5} N/m and the wires repel each other. The current in one wire is 0.58 A. (a) What is the current in the second wire? (b) Are the two currents in the same direction or in opposite direction?

9. In the below figure, switch S_1 is closed while switch S_2 is kept open. The inductance is 270 mH, the resistance is 48.0 Ω and the emf of the battery is 29.0 V. At time t after S_1 is closed, the current in the circuit is increasing at a rate of 7.20 A/s. At this instant what is v_{ab}, the voltage across the resistor?



10. The below figure shows three charges placed on the vertices of a right triangle. The charge q_3 is positive and has a magnitude 3.00×10^{-6} C. Charge q_2 is unknown while the magnitude of charge q_1 is known to be 1.00×10^{-6} C. The sign of q_1 is not known. It is known that the net force **F** on q_3 due to the other two charges is along the negative x-direction. You can take q_1 and q_2 to lie on the x-axis.



a) If $\mathbf{F}_1$ and $\mathbf{F}_2$ are the forces exerted by q_1 and q_2 respectively on q_3, draw a possible diagram depicting $\mathbf{F}_1$ and $\mathbf{F}_2$ such that the net force is **F**.
b) Deduce the signs of q_1 and q_2 based on the scenario obtained in (a). Explain your answer.
c) Calculate the magnitude of q_2.
d) Calculate the magnitude of the net force **F**.